

Effect of Educational Podcasts on Student Performance

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Introduction

Today's students operate in a digital and global environment dominated by digital devices—cell phones, personal digital assistants (PDAs), and mobile devices—and so does today's business (Coleman, 2005). Virtually every electronic device can connect directly to the Internet. Many college students already have access to the Internet through their schools or commercial Internet service providers. Many of today's popular Internet-capable devices include cell phones, laptop computers, PDAs, the iPod, and various digital media players. Digital media players, or iPods, have been very popular among college students because they can serve as an entertainment center (music, video, games, etc.) and a learning device. Many teaching methods can also be enhanced through digital media and podcasting, helping educators become more effective in their teaching. Virtually all course content, lectures, and assignments can easily be converted to MP3 digital format or other audio/video podcast formats, so students can review the course materials outside of the classroom. This is what educational podcasts are all about. According to Fordham University RETC Center for Professional Development, life is about change, and learning through change can be transformative (<http://www.podcastforteachers.org>). Podcasts are about embracing change in teaching and learning, which aligns with transformative learning theory (King, 2005).

Podcasts are derived from Apple's portable music player, the iPod. The iPod and Podcast technologies can deliver educational content to students. Students can play podcasts directly in iTunes (Apple media player) or with a computer media player or any MP3 or digital media player (<http://en.wikipedia.org/wiki/Podcast>).

Podcasting is an audio/video file feed in the Really Simple Syndication (RSS) format, a family of web feed formats used to publish frequently updated digital content, such as blogs. Podcasting is also a term used to describe a collection of technologies for automatically distributing audio and video programs over the Internet. Podcasting enables educators to create self-published audio/video content from course material, lectures, and assignments. Most digital media players or computers with audio-playing software can play podcasts. The term podcast, however, still refers largely to audio/video content distribution (<http://en.wikipedia.org/wiki/Podcast>).

How does podcasting change the way teachers teach and students learn? Podcasts allow a teacher to easily create self-published course content and lectures. A Teacher can also listen to his/her own lectures to improve their presentation skills. Students can download and replay any published podcasts by their teacher that have been previously covered in the classroom. Students can listen to or review any previous lectures outside of the classroom. A Podcast is highly flexible audio/video content that students can listen to at home, on the way to work, or on the go. Listening to podcasts is also a great way to review lectures before an exam.

Podcasts are related to notetaking or reviewing lecture notes. Traditionally, note-taking is a common activity in school settings, and a large percentage of students have been reported to take notes (Palmatier and Bennett, 1974). Cohn, E., Cohn, S., and Bradley (1995) note that when listening and taking notes, students must attend to the lecturer, abstract important information, and then hold and manipulate a representation of that information in working memory while recording notes. In addition, students are less likely to pay attention to the lecture while taking notes. Kiewra (1991) offers suggestions to facilitate the selection

of information to record during a lecture. These suggestions include using cueing strategies such as recording information on the blackboard and repeating the lecture, as well as using a tape or video.

Frey and Birnbaum (2002) have stated that computer-assisted instruction has a positive effect on lectures, especially in taking notes and studying for exams. In addition, research suggests that a technology-based approach to instruction is perceived favorably by students and yields significant improvements in student learning, as evidenced by both student self-reports and objective outcome testing (Smith & Woody, 2000).

The World Wide Web has also created an information-rich teaching and learning environment that facilitates social interaction and exchange among students and instructors. It allows teachers and students to access a wealth of knowledge-based information and assessment tools that replicate and expand the traditional classroom. It has proven effective for storing, disseminating, and retrieving information relevant to the course, and it is available to students anytime, anywhere (Aggarwal & Bento, 2000).

Given this varied set of results, an increasing number of teachers are using technology-assisted instruction in their classes. In this research study, educators will gain a better understanding of students' use of technology provided by their teachers compared with more traditional classroom methods of learning. Educational podcasts are a new technology that allows online audio/video communication between students and teachers, delivered via RSS Podcasting and available for students to listen to on their PDAs or personal computers. Instead of reviewing your class notes, why not replay the class lecture directly on students' portable media players via podcasting? This also comes with a cost for both the students and

the teacher, as they must have an iPod or MP3 player. In addition, teachers must regularly publish podcasts of the class information.

Many studies on technology-based learning have also shown that students are likely to focus on the best learning resources. One such study is by Aytekin (2005), who stated that students learn best when they actively construct their own understanding, as well as through reflective and more responsive forms of education. Educational podcasts are a new technology trend that is widely popular, particularly among students.

One such study examines instructional technologies that support the learning preferences of today's students. Students prefer to use technology to communicate with each other and with their instructor. Communication such as E-mail, discussion boards, instant messaging, blogs, and conference systems have created more efficient and effective learning and teaching environments. With advances in communication, educators have made technology-based learning possible through information technology (Ficek, 2006), including computers and PDAs.

The purpose of this study was to investigate the effect of educational podcasts on student performance (or learning) and determine how podcasts can be used to enhance student performance. More importantly, does the use of podcasts improve student performance? In answering these questions, it is important for us, as educators and researchers, to review any prior related literature on the relationship between podcasts and student performance. If we should in fact find a correlation between podcasts (or technology-based learning) and student performance, we might then be interested in seeking answers to how podcasts can be used to enhance student performance.

According to a previous study by Frey and Birnbaum (2002), technology-based education offers a hands-on approach to improving educational outcomes. Since then, technology has undergone significant advances in delivering educational podcasts via the Internet, digital media players, and iTunes or any other media player software. In an effort to address this new way of teaching and learning via educational podcasts—audio/video podcasts morphed into traditional classroom learning—where lectures, notes, and assignments are podcasted over the Internet to support learning outside the classroom. The important question to address in this study is students’ outcomes. Is the achievement level of students who listen to podcasts greater than students who do not? That is the question addressed in this study.

Method

Participants

In a hypothetical scenario, the participants are to be taken from the normal class size. Half of the students will use podcast-assisted instruction, and the remaining half will use traditional methods and in-class lectures. The demographics of the students in both groups are assumed to be very similar, with the technology-enabled group having access to the Internet and podcasts, in addition to podcast lectures as a supplement to the presentation notes.

Design and Procedure

In the second half of the semester (or term), the teacher poses a prompt or question, and the entire class is given the opportunity to respond. If they would like to use podcast lectures or remain with traditional lecture methods, they can take notes and receive program instructions from the teacher on a class-by-class basis. Depending on familiarity with

podcasts, the sample population will be selected from the normal class size and divided into 2 groups. Group #1 attends class lectures and class discussions. In addition, lecture podcasts will be provided as a supplement to listen to outside of the classroom. Group #2 attends class lectures and follows traditional study methods—note-taking and other study habits. Both groups of students receive the same instruction in the same classroom, with the same teacher leading both groups' classes, including attendance, content, lectures, assignments, and examinations. In addition, all instructions and lectures will be recorded and placed on the course web page for group #1.

The effectiveness of podcasts and students' performance is determined by calculated grades based on homework assignments, class participation, and examinations. The same teacher will assess all students using the same measurement criteria—class attendance, assignments, and examinations (quizzes, mid-term, and final).

Students will be pre-assessed after the first midterm evaluation of performance and post-assessed at the end of the course in the final evaluation. The teacher will examine the calculated grades in both groups and compare the effect of podcasts.

Data for course evaluation will be collected at the end of the class. Students will be asked to respond to questions about their satisfaction with and attitudes toward podcasts. Students will be asked to respond to a question about the effectiveness of podcasts and learning outcomes. A Likert scale of 1-7 will be used (1 = strongly disagree, 7 = strongly agree). Students will be asked to answer about their teacher's preparation, podcast availability, amount of work, etc., with the final question being the instructor's overall effectiveness.

Results

Ideally, based on the calculated grades from the course objectives and learning outcomes, the researcher would like to see evidence that podcasts, as learning tools, can motivate learning, reinforce learning, encourage more fully participating in class, reduce the time spent searching through notes, and ultimately improve academic performance. Based on the previous literature reviews described in this study, it seems important to consider the influence of podcasts and extra time on learning activities, as learning outcomes have been found to influence student academic achievement.

Discussion

Many college instructors are turning to the new and hip way to improve students' grades by podcasting. Podcasts are intended to reinforce the lecture materials. Let us compare the two groups of students: one is the traditional group that attends lectures and takes notes, and the other also attends class and listens to podcasts. Would the second group's grade improve with the use of podcasts? This research study addresses those questions, and based on previous literature reviews and studies, there is strong evidence that the second group outperforms the first when using podcasts to reinforce lecture material. However, there is no such study so far on the effectiveness of podcasts and learning outcomes; there is good reason to believe that students who are willing to listen to podcasts of past lectures are probably not the type of student who skips class for fun. An article in The Boston Globe mentioned a professor from the University of Massachusetts who has been using podcasts for his chemistry class for the last 3 years found that his class' grade point average improved when podcasting was made available (Cristina, 2006). Furthermore, having podcast materials to supplement lectures seems beneficial to students, and having them made available online

for students to download and review becomes an invaluable study aid. At any given point in time, the student has the podcast available for reference. However, more research is needed to address issues related to podcast delivery and its impact on students and society, since new technology like the iPod is so varied and fast-moving that universities and colleges' equipment investments can't be turned over quickly enough to keep pace with rapidly changing technologies.

References

- Aggarwah, A. K., Bento, R. (2000). Web-based education. In A. K. Aggarwal (Ed.), Web-based learning and teaching technologies: Opportunities and challenges, 2- 16.
- Aytekin, I. (2005). A New Model for the World of Instructional Design: A New Model, Turkish Online Journal of Educational Technology, July 2006, Vol. 4, Issue 3
- Coleman M. S. (2005). Testimony to the Senate Subcommittee on Higher Education Appropriations. April 29, 2005, from <http://www.umich.edu/pres/speeches/050429senate.html>
- Ficek, R. (2006, December). Educational Technology; Nov/Dec 2006, Vol. 46 Issue 6, p29-34
- Holley, P. W. (2006). International Journal of Construction Education & Research; Apr2006, Vol. 2 Issue 1, p29-41
- Kreiner, S. K. (1997). Guided Notes and Interactive Methods for Teaching with Videotapes. Department of Psychology, Central Missouri State University, Vol. 24, No. 3
- King, K. P. (2005). Bring Transformative Learning to Life: Understanding its meanings and opportunities for learners and educators. Malabar, FL: Krieger
- Cohn, E., Cohn S., and Bradley J. Jr. (1995). Notetaking, Working Memory, and Learning in Principles of Economics, Journal of Economic Education
- Palmatier, R. A., and J. M. Bennett (1974). Notetaking habits of college students. Journal of Reading 18: 215-18
- Kiewra, K. A., N. F. Dubois, D. Christian, D., McShane A., Meyerhoffer M., Roskelley D. (1991). Note-taking functions and techniques. Journal of Educational Psychology 83:240-45

Smith, S. M., Woody, P. C. (2000). Interactive effect of multimedia instruction and learning styles. *Teaching Psychology*, 27(3), 200-227

Cristina, S. (2006), Podcast craze hits, *The Boston Globe*, from

http://www.boston.com/news/local/articles/2006/07/11/podcast_craze_hits_classrooms/

iPod in Education, “learning on the go”, from <http://www.apple.com/education/products/ipod/>

Wikipedia, the free encyclopedia, <http://en.wikipedia.org/wiki/Podcast>